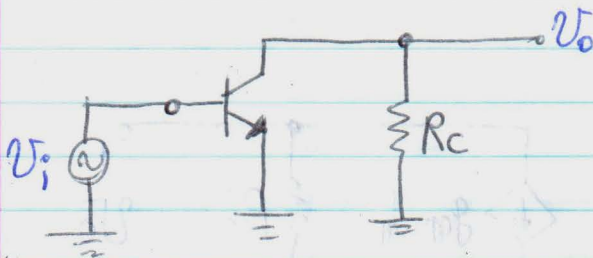


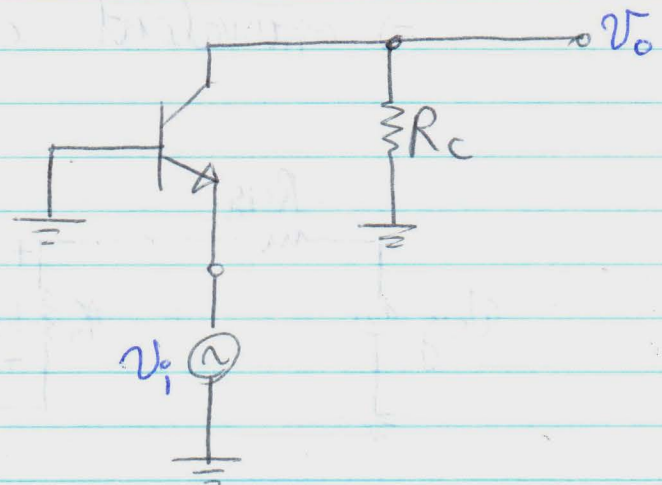
<BJT Basic Configurations>

Almost-linear amplification can be obtained by dc biasing the transistor at an appropriate point in its active region of operation, and by keeping the input signal (V_{be}) small.

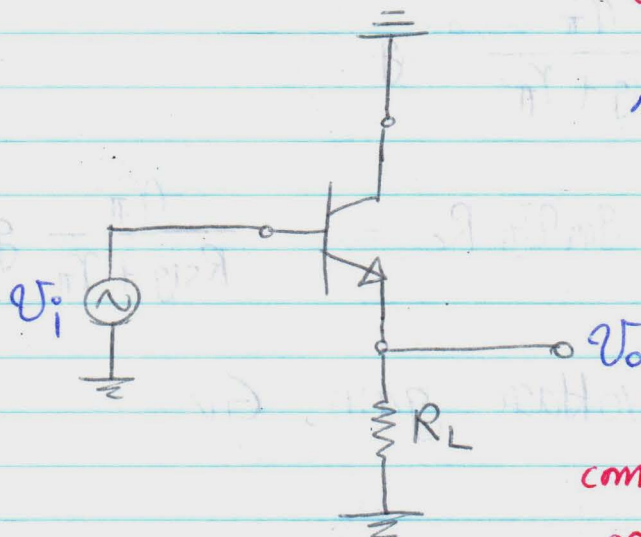
⇒ three basic configurations for a linear amplifier.



Common-Emitter (CE)



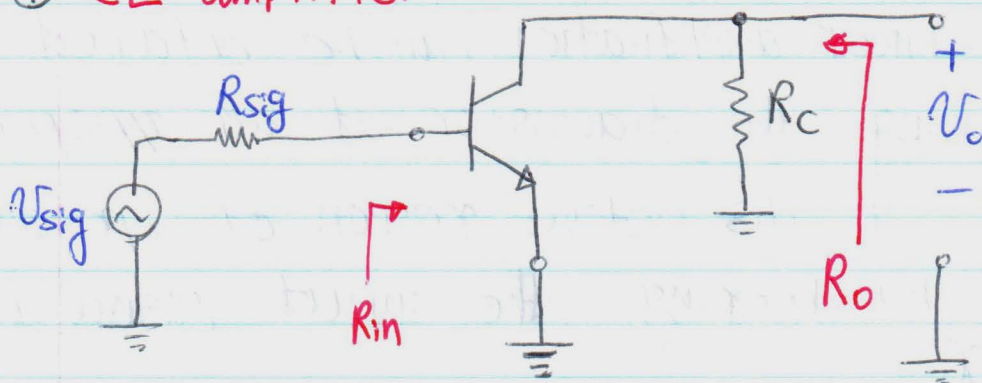
Common-Base (CB)



common-collector (CC)
or Emitter Follower

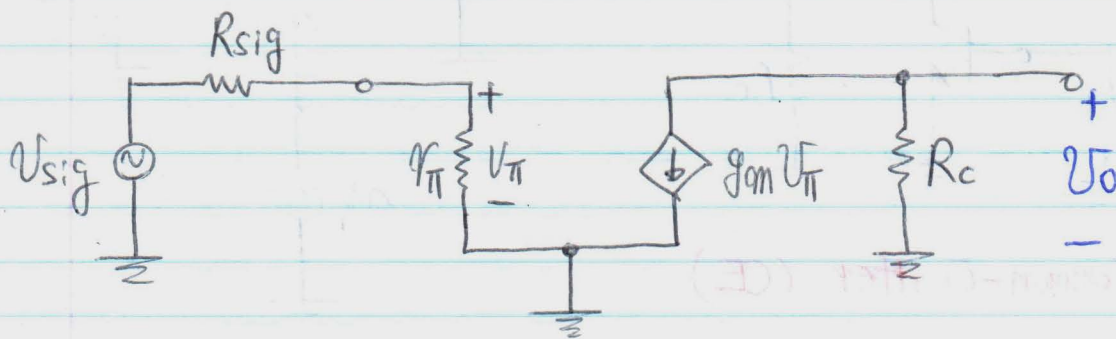
note that bias circuit is not shown here.

① CE amplifier



The bias circuit is omitted.

⇒ equivalent circuit.



$$V_{\pi} = \frac{r_{\pi}}{R_{sig} + r_{\pi}} V_{sig}$$

$$V_o = -g_m V_{\pi} R_C = -\frac{r_{\pi}}{R_{sig} + r_{\pi}} g_m R_C \cdot V_{sig}$$

⇒ Overall voltage gain, G_v

$$G_v \equiv \frac{v_o}{v_{sig}} = - \underbrace{\frac{r_{\pi}}{r_{\pi} + R_{sig}}}_{\text{voltage divider}} g_m \cdot R_c$$

- the effect of the finite input resistance (r_{π})
 $\Rightarrow$ reducing G_v by the voltage-divider ratio of $r_{\pi} / (r_{\pi} + R_{sig})$.

- to maximize G_v ,
 $R_{sig} \ll r_{\pi}$.

then, $G_v \simeq -g_m R_c$

- CE amplifier.

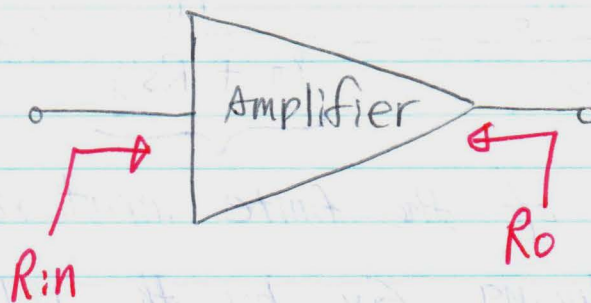
- the most useful of all transistor amplifier configurations.

- input resistance of CE amplifier, R_{in} ,

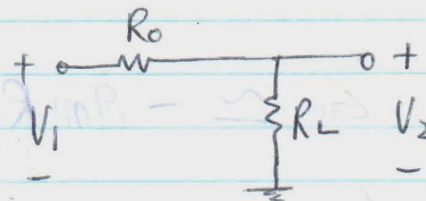
$$R_{in} = r_{\pi} = \frac{\beta}{g_m} = \frac{\beta}{I_c} V_T$$

if $I_c \downarrow \rightarrow R_{in} \uparrow$ but $g_m \downarrow$, therefore, the G_v is also lowered.

$\Rightarrow$ this is a design trade-off.

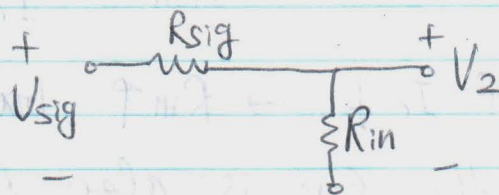


- the output resistance R_o should be much smaller than the load resistance R_L when the amplifier is connected to a load.



when $R_o = 0$,
 $V_1 = V_2$

- the input resistance R_{in} should be much greater than the resistance of the signal source when an input signal is connected to the amplifier.

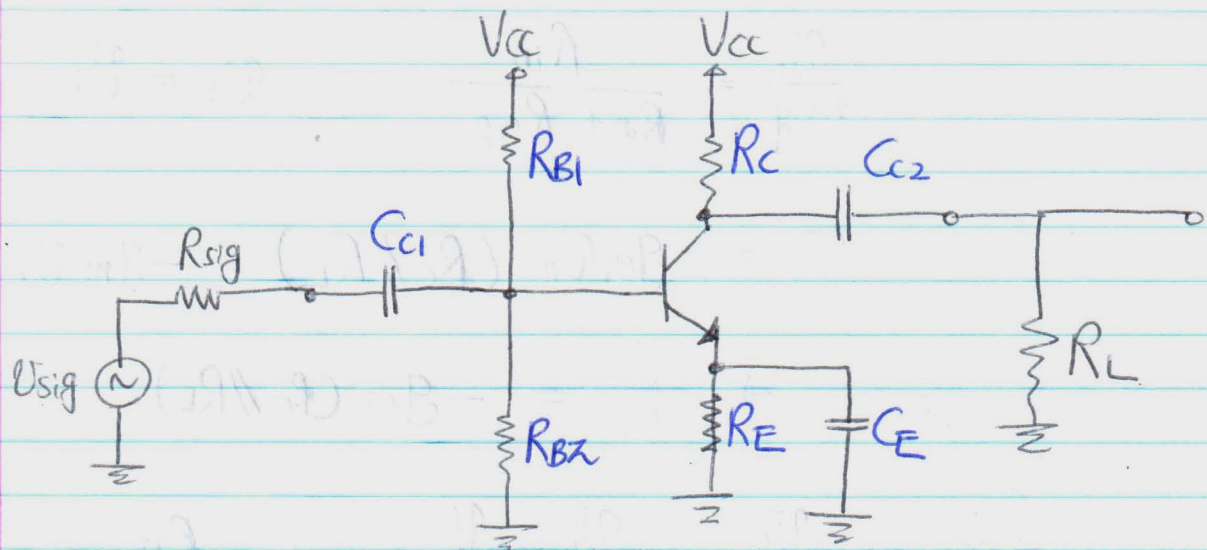


- CE amplifier

$$R_{in} = r_{\pi}$$

$$R_o = R_c$$

④ CE amplifier with a bias circuit.



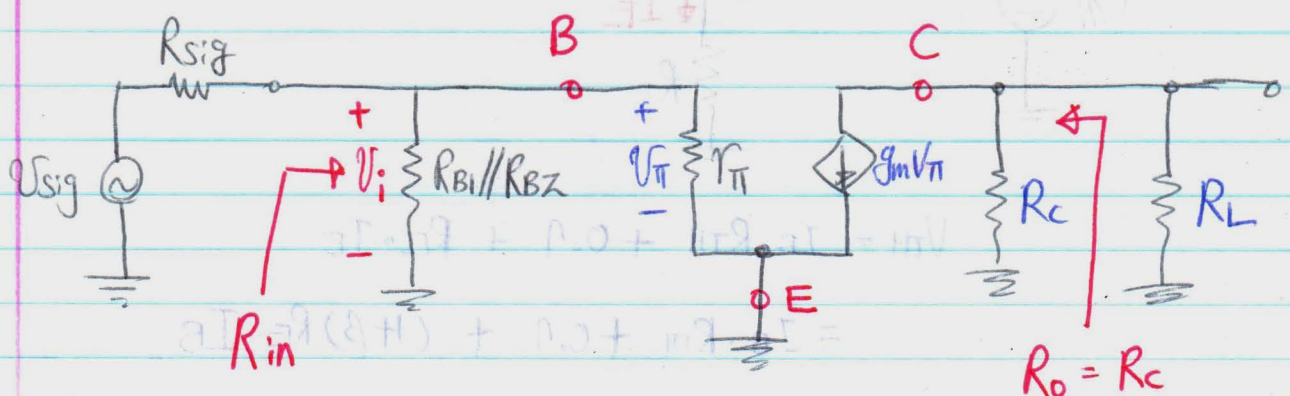
• C_{c1}, C_{c2} : coupling capacitor :

make a short circuit at all signal frequencies of interest while blocking dc.

$\Rightarrow$ preserving dc-bias.

• C_E : bypass capacitor = signal ground.

• equivalent circuit



$$R_{in} = R_{B1} // R_{B2} // r_{\pi}$$

$$\frac{V_i}{V_{sig}} = \frac{R_{in}}{R_{in} + R_{sig}}, \quad V_{\pi} = V_i$$

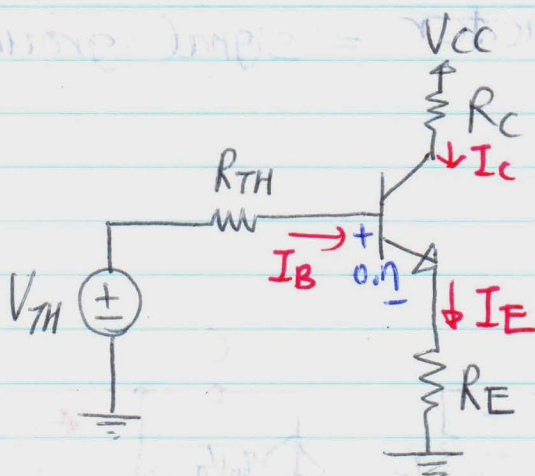
$$V_o = -g_m V_{\pi} (R_c // R_L) = -g_m V_i (R_c // R_L)$$

$$\Rightarrow \frac{V_o}{V_i} = -g_m (R_c // R_L)$$

$$\Rightarrow G_v \equiv \frac{V_o}{V_{sig}} = \frac{V_i}{V_{sig}} \times \frac{V_o}{V_i} = - \frac{R_{in}}{R_{in} + R_{sig}} g_m (R_c // R_L)$$

$$\text{where } R_{in} = R_{B1} // R_{B2} // r_{\pi}$$

• DC-bias circuit.



$$V_{TH} = \frac{R_{B2}}{R_{B1} + R_{B2}} V_{CC}$$

$$R_{TH} = R_{B1} // R_{B2}$$

$$V_{TH} = I_B \cdot R_{TH} + 0.7 + R_E \cdot I_E$$

$$= I_B \cdot R_{TH} + 0.7 + (1 + \beta) R_E \cdot I_B$$

$$\Rightarrow I_B = \frac{V_{TH} - 0.7}{R_{TH} + (1 + \beta)R_E}$$

$$I_C = \beta I_B$$

$$g_m = \frac{I_C}{V_T}, \quad r_\pi = \frac{\beta}{g_m}$$